

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“AI Automation System”

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

BACHELOR OF ENGINEERING IN
Department of Artificial Intelligence and Machine Learning

Submitted by:

POOJA SINGARI - 1DS20AI040

Conducted at



Seed Formulai

Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning



CERTIFICATE

This is to certify that the Internship titled “**AI Automation System**” carried out by **Ms POOJA SINGARI [1DS20AI040]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (**20AI8ICINT**)

Signature of Reporting Head
Mala S
Founder
Seed Formulai

Signature of Internship Coordinator
Prof. Kavya D N
Dept. of AI & ML
DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

1) _____

2) _____

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **Pooja Singari**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College Of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **Seed Formulai**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023-2024.

Date:

USN: 1DS20AI040

Place: Bangalore

NAME: POOJA SINGARI

OFFER LETTER



SEED FORMULAI

January 29, 2024

Subject: Internship Offer Letter

Dear Pooja Singari,

We are pleased to offer you the position of AI Intern at Seed Formulai with a start date of 05/02/24, contingent upon a background check. You will be reporting directly to Jay at Whitefield location. We believe your skills and experience are an excellent match for our company.

In this role, you will be required to engage in various activities including data analysis, model development, and contributing to our AI projects. You will work closely with senior team members to enhance our AI capabilities and deliver innovative solutions.

Throughout your internship, you will have access to comprehensive training and development programs designed to enhance your professional skills and knowledge in artificial intelligence and data analysis.

Your employment with Seed Formulai will be on an at-will basis, which means you and the company are free to terminate the employment relationship at any time. This letter is not a contract or guarantee of employment for a definite amount of time.

As an intern of Seed Formulai, you are eligible for our benefits program, which includes performance-based incentives and other benefits which will be described in more detail in the orientation package.

Please confirm your acceptance of this offer by signing and returning this letter by 02/02/24. We are excited to have you join our team! If you have any questions, please feel free to reach out at any time.

Sincerely,

Founder

Accept Terms:

Signature with Date:

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Mrs. Mala S**, Seed Formulai, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, **Prof. Kavya D N**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

POOJA SINGARI [1DS20AI040]

TABLE OF CONTENTS

Sl no	Description	Page no
1	INTRODUCTION	1-2
2	DESCRIPTION OF ORGANIZATION	3-4
3	SCOPE OF WORK	5-6
4	LEARNING OBJECTIVES AS GIVEN BY COMPANY	7
5	CHALLENGES AND SOLUTIONS	8-10
6	ACHIEVEMENTS AND CONTRIBUTION	11-12
7	REFLECTION AND ANALYSIS	13
8	RECOMMENDATIONS	14
	CONCLUSION	15
	REFERENCES	16

CHAPTER 1

INTRODUCTION

INTRODUCTION TO AI AUTOMATION SYSTEM

An automation system refers to a combination of hardware and software components designed to perform tasks automatically that would otherwise require human intervention. Automation systems powered by artificial intelligence (AI) are revolutionizing how enterprises operate, compete, and thrive in today's fast-paced market environments. By integrating AI-driven technologies, businesses are able to automate complex and labor-intensive processes, transforming both their operational efficiency and strategic capabilities. These systems enable companies to execute tasks with unprecedented speed and precision, reduce human error, and manage data-driven decisions with ease. As a result, enterprises can significantly enhance productivity, improve customer satisfaction, and maintain competitive advantages in their respective industries.

1. Operational Efficiency

Automation of routine and repetitive tasks frees up human resources to focus on more strategic initiatives. Systems like Robotic Process Automation (RPA) can handle tasks such as data entry, scheduling, and report generation, which speeds up workflows and reduces overhead costs.

2. Enhanced Decision-Making

Machine learning algorithms can analyze large volumes of data to detect patterns and predict future trends. This capability allows enterprises to make informed decisions quickly, improving areas like inventory management, financial forecasting, and customer service.

3. Improved Customer Experience

Natural Language Processing (NLP) enables more sophisticated customer interaction tools, such as chatbots and virtual assistants. These tools can provide immediate, 24/7 customer service, improving response times and customer satisfaction, which are vital for retention and growth.

4. Quality Control and Compliance

AI-driven automation systems can monitor and ensure compliance with regulations and standards without human oversight. In sectors like manufacturing, computer vision technologies can identify defects and irregularities in products, ensuring high quality and reducing waste.

5. Scalability

Automation systems allow businesses to scale operations without a proportional increase in costs. This scalability is especially crucial for handling growth phases, such as market expansions or product launches, without compromising service quality or operational speed.

By leveraging automation systems, enterprises not only streamline their internal processes but also enhance their ability to adapt to market changes and consumer demands, ultimately supporting sustained business growth.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

SEED FORMULAI



Introduction

Seed Formulai is at the forefront of transforming early-stage startups by leveraging cutting-edge data and technology to fuel growth and enhance operational efficiency. This innovative company is dedicated to redefining the landscape of business development through its comprehensive solutions, tailored specifically for companies eager to capitalize on data-driven insights. With a strong emphasis on guiding product development, market strategies, and operational choices, Seed Formulai equips its clients with the tools and expertise needed to navigate the complexities of today's business environment. The company's commitment to innovation and progress makes it an indispensable partner for startups aiming to make a significant impact in their respective industries.

Founding Team

Seed Formulai is powered by a founding team with a wealth of experience across the global tech and business landscape. This seasoned group of professionals brings together decades of expertise from working in diverse markets and industries worldwide. They have not only been pivotal in driving technological advancements and implementing robust data strategies in multiple countries but also have a rich history of entrepreneurial success.

Each member of the founding team has been strategically selected for their unique skill sets and deep insights into the mechanics of business growth, data utilization, and technology deployment. Their combined experience spans across key sectors including tech startups, data science, business intelligence, and corporate strategy, providing Seed Formulai with a solid foundation to support new ventures.

ABOUT THE COMPANY

About Seed Formulai

VISION STATEMENT

To be the catalyst for transformative growth in startups worldwide, setting the standard for innovation and efficiency in the digital age.

MISSION STATEMENT

Mission at Seed Formulai is to empower early-stage companies to achieve their fullest potential by providing advanced data-driven tools and expert guidance.

Optimize Operational Efficiency: Streamline processes and increase productivity to ensure efficient business operations.

Provide Advanced Data-Driven Tools: Utilize cutting-edge technology to deliver actionable insights that inform critical business decisions.

Drive Strategic Market Advancement: Help clients identify and exploit market opportunities for sustainable growth.

CHAPTER 3

SCOPE OF WORK

During my internship at Seed Formulai, I was involved in several high-impact projects, each representing a distinct component of an automation system designed to address complex business challenges through advanced technology.

1. Enhanced Multi-label Text Classification System:

Objective: Elevate the precision and dependability of a system that categorizes text into multiple, overlapping categories.

Scope:

- Data Optimization: Implement sophisticated imputation methods to handle significant data gaps and employ advanced natural language processing techniques to prepare the text data.
- Model Innovation and Assessment: Utilize leading machine learning frameworks to craft models adept at managing the nuances of multi-label data. Leverage synthetic datasets and cross-validation to ensure robust model evaluation in the absence of a conventional test set.

2. Ethically-Guided Large Language Model (LLM) Chatbot:

Objective: Create a responsive chatbot powered by large language models, designed to interact effectively while adhering to ethical standards.

Scope:

- Data Handling: Apply cutting-edge algorithms for precise text embedding and employ techniques to sustain dialogue flow and context.
- Integration of Ethical Guardrails: Implement a dual approach of rule-based and machine learning mechanisms to enhance the chatbot's contextual understanding and prevent inappropriate responses.

3. Automated GitHub Repository Analysis:

Objective: Automate the monitoring and analysis of GitHub repositories to pinpoint and assess influential software development projects.

Scope:

- **Data Management:** Develop automation scripts to fetch data via Snowflake, identify top repositories by star ratings, and maintain an updated list.
- **Automation of Communications:** Utilize email and Google Calendar APIs via Zapier to streamline updates and manage reminders for regular repository reviews.

Each project was designed to not only address specific technical challenges but also to advance Seed Formulai's capabilities in data-driven decision-making and operational efficiency. This structured scope of work allowed me to significantly contribute to the company's objectives while enhancing my skills in data science, machine learning, and software automation.

CHAPTER 4

LEARNING OBJECTIVES

The learning objectives set by the company were designed to foster a comprehensive development of skills in several key areas of data science and artificial intelligence:

1. Proficiency in Machine Learning and Deep Learning:

Develop a strong foundation in both traditional machine learning techniques and modern deep learning architectures. This includes building predictive models and TensorFlow for creating more complex neural network-based applications. Gain deep insights into NLP techniques to handle and process text data efficiently. This includes using state-of-the-art tools and platforms to develop applications like chatbots that can understand and generate human-like text responses.

2. Advanced Data Handling and Visualization:

Enhance abilities in managing, analyzing, and visualizing data effectively using a variety of tools. This includes learning to implement data warehousing solutions with platforms like Snowflake, and utilizing visualization libraries such as Seaborn to derive actionable insights from data.

3. Automation and Integration of Technologies:

Acquire skills in automating processes and integrating various APIs and tools to create efficient workflows. This involves using automation platforms like Zapier to connect disparate systems and streamline operations, enhancing productivity and reducing manual effort.

4. Adaptability to AI Advancements:

Develop the skill to swiftly adapt to emerging advancements in artificial intelligence, particularly in the realm of generative AI and large language models (LLMs). This involves keeping abreast of the latest developments in the AI field, understanding the capabilities of new generative technologies, and quickly mastering new tools and approaches to effectively integrate them into existing systems.

CHAPTER 5

CHALLENGES AND SOLUTIONS

During my internship at Seed Formulai, I was involved in three major projects, each aimed at solving specific challenges and improving processes through innovative technological solutions:

1. Advanced Multi-label Classification System:

Challenges:

- **Data Integrity:** The dataset was plagued by a high volume of missing values, risking the accuracy of model training.
- **Lack of Test Data:** The absence of a dedicated test dataset made it challenging to assess model performance accurately.
- **Raw Text Handling:** Processing and cleaning raw text data to make it suitable for analysis was essential yet challenging.

Solutions:

- **Robust Data Imputation:** Applied various advanced imputation methods to manage missing values and sometimes used predictive modeling for data replacement.
- **Validation Strategy:** Implemented cross-validation techniques to validate the model's effectiveness in a real-world scenario.
- **Sophisticated Text Processing:** Utilized comprehensive NLP techniques, including tokenization, stemming, and stopwords removal, to refine the input text data.

2. Ethical LLM Chatbot Development:

Challenges:

- **Effective Data Embedding:** Ensuring accurate embedding of data and maintaining dialogue context without disruption.
- **Contextual Relevance:** The need for the chatbot to sustain context and avoid irrelevant or inappropriate responses was critical.

Solutions:

- **Enhanced Data Management:** Developed precise algorithms for data embedding and sophisticated techniques for text splitting to preserve dialogue flow.

- **Guardrails for Ethical Interaction:** Improved the chatbot's ability to understand context with advanced training and integrated both rules-based and algorithmic guardrails to manage and mitigate out-of-context responses.

3. GitHub Repository Analysis Automation:

Challenges:

- **Popularity-Based Sorting:** Identifying and focusing on the most-starred repositories required efficient data sorting techniques.
- **Streamlined Communication:** Establishing automated processes for periodic communications and updates was necessary.

Solutions:

- **Automated Data Extraction and Ranking:** Crafted a script for dynamic querying of GitHub's API to continuously extract and update a ranked list of repositories.
- **Automated Alerts and Scheduling:** Implemented systems that integrate with email and Google Calendar APIs to automate the dissemination of updates and manage reminders for regular reviews, ensuring proactive project monitoring.

These projects not only addressed significant technical challenges but also enhanced my ability to implement and manage advanced solutions in real-world applications, furthering Seed Formulai's goals in technological innovation and efficiency.

HARDWARE AND SOFTWARE REQUIREMENTS

Software requirements:

- Data Base: Snowflake, Pinecone
- API: Fast-API
- Frameworks: TensorFlow, LlamaIndex, Langchain

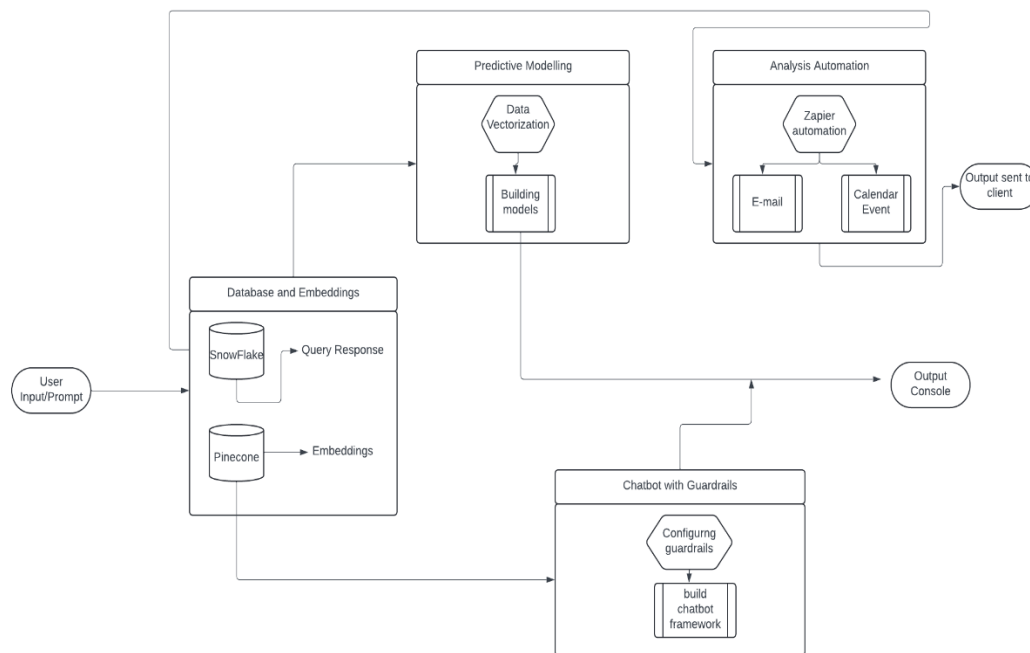
Hardware Requirements:

- Processor: Intel core i5 processor
- Memory: 16 GB
- Hard Disk: 1TB SSD
- GPU: NVIDIA GeForce MX250

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

System Design:



The Architecture of the AI Automation System:

1. User Input/Prompt

Users interact with the system by providing input or prompts, which are then fed into different branches of the system, depending on the nature of the request.

2. Database and Embeddings

- **Snowflake:** This component acts as a data warehousing solution, storing and organizing data efficiently. It receives queries and returns responses for processing.
- **Pinecone:** Utilized for handling vector embeddings, it facilitates the retrieval of similar items or documents by transforming text into vector space, which is essential for operations like semantic search or similarity matching in the chatbot framework.

3. Predictive Modelling

- Data Vectorization: Converts raw data into a numerical format that machine learning models can understand and process.
- Building Models: Involves the construction and training of predictive models that can classify input data or predict outcomes based on the vectorized data.

4. Chatbot with Guardrails

Sets up constraints and ethical guidelines to which the chatbot must adhere, preventing it from generating harmful or inappropriate content by utilizing the embeddings from Pinecone for contextual understanding and response generation.

5. Analysis Automation

- Zapier Automation: Orchestrates automated tasks that involve integrating different applications and services, such as sending emails and creating calendar events based on triggers or conditions.
- E-mail, Calendar Event: Automated emails are configured to be sent to stakeholders or clients with information as well as scheduling reminders on Calendars for timely review.

6. Output Console

All processed are ultimately delivered to an output console. This may be a dashboard, an application interface, or a report that is then sent to the client.

Flow of the Process : Input Prompt > Database and Embedding > Predictive Modelling > > Chatbot with Guardrails > Analysis Automation > Output Console

During my tenure at Seed Formulai, my involvement in distinct projects contributed significantly to the development of a comprehensive automation system aimed at enhancing business efficiency and decision-making. I was instrumental in creating a comprehensive Automation System that significantly enhanced organizational efficiency through intelligent data processing. I crafted a system that not only streamlined reporting and analysis but also delivered tailored insights and managed data with unprecedented accuracy. Collectively, these contributions not only fortified individual project outputs but also synergistically enhanced the functionality and robustness of the automation system.

CHAPTER 7

REFLECTION AND ANALYSIS

Reflecting on my internship at Seed Formulai, I've had the enriching opportunity to engage with various challenging and innovative projects, ranging from machine learning applications to natural language processing and automation tools. Each project, whether it was about enhancing classification systems, developing ethically-guarded AI, or streamlining data analysis from software development platforms, presented unique challenges that broadened my technical and analytical skills.

Throughout my time at Seed Formulai, I have grown significantly in my ability to handle complex data-driven problems and implement practical solutions. The environment fostered here is one of constant learning and adaptation, encouraging not only technical proficiency but also a strong ethical consideration in how technology is applied. The collaborative nature of the work and the mentorship from experienced professionals have profoundly impacted my professional development, enhancing my problem-solving capabilities and my understanding of technology's role in business optimization.

Additionally, working in the fast-paced, agile environment of a startup like Seed Formulai has sharpened my abilities in rapid problem-solving and decision-making. The necessity to swiftly adapt to new challenges and continuously iterate on solutions has not only accelerated my technical learning curve but also enhanced my resilience and flexibility. This experience has instilled in me a keen ability to thrive under pressure and a profound appreciation for the collaborative, innovative spirit that drives startup culture.

Overall, my internship experience has been incredibly formative, cementing my passion for data science and AI, and preparing me well for future endeavors in the tech industry. The skills and insights gained at Seed Formulai are invaluable, and I look forward to applying them in my continued career growth.

CHAPTER 8

RECOMMENDATIONS

- 1. Enhance Cross-Departmental Collaboration:** Develop structured programs that encourage interaction and project collaboration across different departments. Initiating regular cross-functional meetings and team-building activities can foster a culture of open communication and idea sharing. This approach can lead to more holistic solutions to business challenges and faster innovation cycles.
- 2. Implement a Comprehensive Mentorship Program:** Establish a mentorship program that pairs less experienced employees with senior leaders. This initiative should not only focus on skills development but also on fostering leadership qualities and a deeper understanding of the company's strategic goals. Encouraging mentors to work across departments can help disseminate diverse skills and knowledge, enriching the professional growth of mentees and strengthening the overall competency of the workforce.
- 3. Create Innovation Hubs:** Set up dedicated spaces where employees from various departments can come together to brainstorm and experiment with new ideas. These innovation hubs should be equipped with the latest tools and technologies, allowing teams to prototype and test solutions rapidly, thus accelerating the development of new products and services.
- 4. Invest in Emerging Technologies:** Continuously scout and invest in cutting-edge technologies, especially in the fields of AI and machine learning. This could involve partnerships with tech startups or dedicated R&D efforts to integrate advancements like quantum computing or blockchain for enhanced data security and processing capabilities.

CHAPTER 9

CONCLUSION

In conclusion I can affirm that the experience as an intern at Seed Formulai has been profoundly enriching and instrumental in shaping my professional trajectory. Engaging in complex projects such as the multi-label classification, the LLM chatbot with guardrails, and the automation of GitHub repository analysis, I was exposed to real-world applications of advanced technologies and challenged to deliver practical solutions under professional guidance.

The hands-on experience with tools like TensorFlow, OpenAI, and Snowflake, among others, has significantly bolstered my technical expertise. The projects allowed me to navigate through the nuances of machine learning, natural language processing, and data automation, enhancing my problem-solving skills and my ability to innovate within ethical boundaries.

Moreover, the startup environment at Seed Formulai, known for its fast-paced and innovation-driven culture, pushed me to adapt quickly, think critically, and work efficiently within teams. This not only improved my technical abilities but also my soft skills, including communication, teamwork, and project management.

Overall, this internship has been a cornerstone experience, confirming my passion for data science and technology while providing a solid foundation for my career. I am grateful for the mentorship received and the opportunities to contribute meaningfully to Seed Formulai's goals. I look forward to carrying forward these learnings and experiences into my future endeavors in the tech industry.

CHAPTER 10

REFERENCES

Founder: Mala

E-mail: admin@seedformulai.com